CSC311: Introduction to Machine Learning

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Contents

1 Introduction **1**
1.1 Input Vectors . . . 2
1.2 Nearest Neighbours . . . 2

2 Decision Trees **4**
2.1 Entropy . . . 5
2.2 Information Gain . . . 6
2.3 Decision Tree Construction . . . 7

3 Generalization and Bias-Variance Decomposition **8**
3.1 Generalization . . . 8
3.2 Bias-Variance Decomposition . . . 9
3.3 Bagging . . . 11

4 Linear models **12**
4.1 Introduction . . . 13
4.2 Vectorization . . . 13
4.3 Optimization . . . 14
4.4 Feature Mapping . . . 15

A Probability **16**

B Linear Algebra **17**

1 Introduction

For many problems, such as recognizing people and objects, and understanding human speech, it is difficult to program the correct behaviour by hand. In other cases, systems need to adapt to a changing environment (e.g. spam detection), or that we want the system to perform better than human programmers, or even for privacy or fairness reasons. A machine learning algorithm automatically learns from data or experience.

Machine learning and statistics rely on similar mathematics but they differ in their goals. Statistics is more concerned with helping scientists and policymakers draw good conclusions,

whereas machine learning is more concerned with building autonomous agents. Statistics puts more emphasis on interpretability and mathematical rigor, whereas machine learning puts on emphasis on predictive performance, scalability and autonomy.

Types of machine learning include:

Supervised learning: The system takes in inputs and labels/targets of the correct behaviour

Unsupervised learning: The system takes in inputs and look for "interesting" patterns in the data

Reinforcement learning: A learning system (agent) interacts with the world and learns to maximize a scalar reward signal

For much of this set of notes, we focus on supervised learning.

Examples of machine learning include computer vision, speech to text, natural language processing (machine translation, sentiment analysis, topic modeling, spam filtering), e-commerce and recommender systems, etc.

1.1 Input Vectors

Machine learning algorithms need to handle lots of types of data: images, text, audio waveforms, etc. Hence the common stategy is to represent the input as an ***input vector*** in $\mathbb{R}^d$. The mapping to another space that is easy to manipulate is known as ***representation***.

Mathematically, the training set consists of a collection of pairs of an input vector $\mathbf{x} \in \mathbb{R}^d$ and its corresponding ***target***, or ***label***, t:

Regression: t is a real number (e.g. stock price)

Classification: t is an element of a discrete set $\{1, ..., C\}$

We denote the traning set $\{(\mathbf{x}^{(1)}, t^{(1)}), ..., (\mathbf{x}^{(N)}, t^{(N)})\}$

1.2 Nearest Neighbours

We start our discussion of machine learning algorithms with ***nearest neighbours***, which is a quasi-machine learning algorithm, or we say instance-based algorithm, as there is no "learning" involved.

Suppose we are given a novel input vector $\mathbf{x}^*$ to classify. The idea is to find the nearest input vector to $\mathbf{x}$ in the training set and copy its label. We can formalize "nearest" in terms of Euclidean distance

$$||\mathbf{x}^{(a)} - \mathbf{x}^{(b)}||_2 = \sqrt{\sum_{j=1}^{d} (x_j^{(a)} - x_j^{(b)})^2}$$

Algorithm (NN):
1. Find $(\mathbf{x}^*, t^*)$ from the training set closest to $\mathbf{x}$:

$$\mathbf{x}^* = \underset{\mathbf{x}^{(i)} \in \text{training set}}{\text{argmin}} \quad \text{distance}(\mathbf{x}^{(i)}, \mathbf{x})$$

2. Output $y = t^*$

A ***Voronoi diagram*** is a type of geometric diagram that partitions a plane into regions based on the distance to a set of points called the seeds or generators. In a Voronoi diagram, each region is defined as the set of all points in the plane that are closer to a particular seed point than any other seed point. The boundaries between the regions are formed by lines that are equidistant between adjacent seed points. The ***decision boundary*** is the boundary between regions of input space assigned to different categories.

However, the method of nearest neighbours is sensitive to noise or mislabeled data ("class noise"). We therefore smooth the noise by having k nearest neighbours vote.

Algorithm (kNN):
1. Find k pairs $\{\mathbf{x}^{(i_1)}, t^{(i_1)}\}, \{\mathbf{x}^{(i_2)}, t^{(i_2)}\}, ..., \{\mathbf{x}^{(i_k)}, t^{(i_k)}\}$
2. Output the majority class

$$\mathbf{y}^* = \underset{t^{(z)} \in \text{class labels}}{\text{argmax}} \quad \sum_{i=1}^{k} \delta(t^{(z)} = t^{(i)})$$

There are tradeoffs whether we choose small k or large k. Small k is good at capturing fine-grained patterns but may ***overfit***, i.e. be sensitive to random idiosyncracies in the training data. Large k makes stable predictions by averaging lots of examples but may ***underfit***, i.e. fail to capture important regularities. The optimal choice of k depends on number of data points n. We get "nice theoretical properties" if $k \to \infty, \frac{k}{n} \to 0$, i.e. as the number of neighbors considered becomes much larger compared to the size of the training set, the model's bias increases and the variance decreases (the discussion of which we will leave to a later section). In other words, the model becomes less influenced by individual data points and more driven by the overall distribution and structure of the data. The rule of thumb is to choose $k < \sqrt{n}$. But if n is big, then we have a lot of k's to choose from. We can choose k using a ***validation set***.

We would like our algorithm to generalize to data it hasn't seen before. A ***test set*** can be used to measure the ***generalization error*** (error rate on new examples).

A ***hyperparameter*** is a parameter whose value is set before the learning process begins and remains constant during training. Unlike model parameters, which are learned from the training data, hyperparameters are typically chosen by the practitioner or determined through a process called hyperparameter tuning. k is a hyperparameter in this case.

In Figure 1, the test error of the training set is highest when k is highest and lowest when k is lowest, as it is fitting to the noise. In contrast, for the test set, there is a minimum test error at a certain value of k. We therefore choose this k for our model. The set of data used for this purpose is often called the validation set, and we typically split our data into three portions —

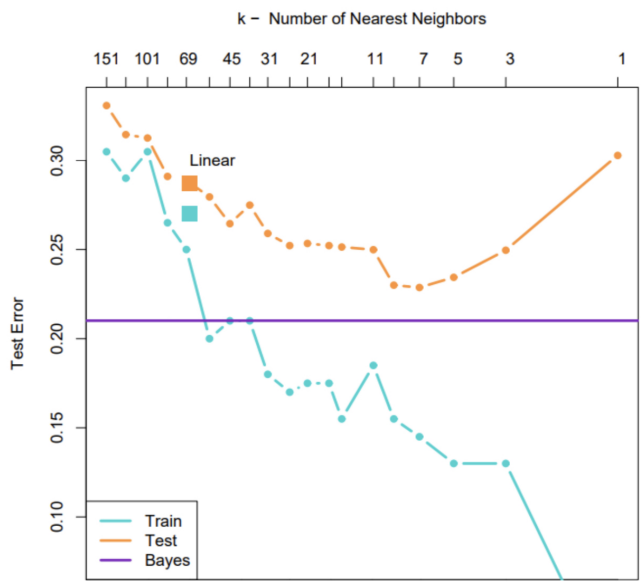

Figure 1: Test error in training set and test set with decreasing k

training set, validation set, and test set. The ideal model has low training error and test error.

Even if we chosen our k well, the method of k-Nearest Neighbours still has certain pitfalls:

i. **The Curse of Dimensionality.** High dimensional data can have negative impact on the performance and effectiveness of the algorithm.

   Increased data sparsity: The volume of a single ball of radius ϵ around each point is $\mathcal{O}(\epsilon^d)$. The total volume of $[0,1]^d$ is 1. Therefore $\mathcal{O}((\frac{1}{\epsilon})^d)$ points are needed to cover the volume, making it more difficult to identify neighboring points that are close in proximity. The reduced density makes it challenging to define a clear neighborhood around a given data point.

   A saving grace is that some datasets may have low ***intrinsic dimension***, i.e. lie on or near a low-dimensional manifold.

ii. **Computation Cost.** The number of computations at training time is 0, but the number of computations at test time per query is

   * Calculate D-dimensional Euclidean distance with N data points: $\mathcal{O}(ND)$
   * Sort the distances: $\mathcal{O}(N \log N)$

   which is very expensive by standards of a learning algorithm. Also, we need to store the entire dataset in memory.

iii. **Normalization** ...

2 Decision Trees

A deicision tree is a tree with internal nodes that test a feature, branches determined by the feature value, and leaf nodes that represent outputs / final predictions. Thinking of the decision

tree as partition Each path from the root to a leaf defines a region R_m of input space. We denote the training examples that fall into R_m to be $\{(x^{(m_1)}, t^{(m_1)}), ..., (x^{(m_1)}, t^{(m_1)})\}$.

For regression trees, the output is continuous, and the leaf value y_m is typically set to the mean value in $\{t^{(m_1)}, ..., t^{(m_k)}\}$. For classification trees, the output is discrete, and the leaf value y_m is typically set to the most common value in $\{t^{(m_1)}, ..., t^{(m_k)}\}$.

Decision trees are universal function approximators, meaning that they can approximate any arbitrary function with a given level of accuracy. Finding the smallest decision tree that correctly classifies a training set is NP-complete.[1] We therefore resort to a ***greedy heuristic***:

* Start with the whole training set and an empty decision tree
* Pick a feature and candidate split that would most reduce a loss (a scalar number to assess whether we are on track, the lower the loss the better the split.)
* Split on that feature and recurse on subpartitions

2.1 Entropy

The ***entropy*** of a discrete random variable is the uncertainty inherent in its possible outcomes, with unit bit. The system whose outcomes are more certain has lower entropy (e.g. A loaded coin has a lower entropy than a fair coin). We can also think of entropy as the expected information content of a random draw

You cannot store the outcome of a random draw using fewer expected bits than the entropy without losing information.

The ***entropy*** of a discrete random variable Y is given by

$$H(Y) = -\sum_{y \in Y} p(y) \log_2 p(y)$$

For multivariables X and Y,

$$H(X, Y) = -\sum_{y \in Y} \sum_{x \in X} p(x, y) \log_2 p(x, y)$$

Hence we also have the conditional entropy

$$H(Y|X = x) = -\sum_{y \in Y} p(y|x) \log_2 p(y|x)$$

where we used $p(x|y) = \frac{p(x,y)}{p(x)}$ and $p(x) = \sum_y p(x, y)$

[1] "Nondeterministic polynomial-time completeness." (NP-completeness) is a concept in computational complexity theory that classifies computational problems based on their difficulty and solvability. A computational problem is considered NP-complete if it belongs to the class of NP problems and has the property that any other problem in NP can be reduced to it in polynomial time. In other words, an NP-complete problem is one for which a solution can be verified in polynomial time, and any problem in NP can be transformed into it efficiently. The significance of NP-completeness lies in the fact that if a problem is proven to be NP-complete, it implies that solving that problem efficiently (in polynomial time) is unlikely. In other words, no efficient algorithm is known to solve all instances of an NP-complete problem.

Also the expected conditional entropy

$$\mathbb{E}[H(Y|X)] = \mathbb{E}_x[H(Y|X=x)] = \sum_{x \in X} p(x) H(Y|X=x) = -\sum_{x \in X} \sum_{y \in Y} p(x,y) \log_2 p(y|x)$$

Some useful properties of entropy include

* H is always non-negative
* Chain rule: $H(X,Y) = H(X|Y) + H(Y) = H(Y|X) + H(X)$
* If X and Y are independent, then X does not affect our uncertainty about Y: $H(Y|X) = H(Y)$
* Knowing Y makes our knowledge of Y certain: $H(Y|Y) = 0$
* Knowing X decreases only uncertainty about Y: $H(Y|X) \leq H(Y)$

High entropy indicates:

* Variable has a uniform like distribution over many outcomes
* Flat histogram
* Value sampled from it are more predictable

Low entropy indicates:

* Distribution is concentrated on only a few outcomes
* Histogram is concentrated in a few areas
* Values sampled from it are more predictable

2.2 Information Gain

Given two random variables Y and X, the ***information gain*** in Y due to X, or the ***mutual information*** of Y and X is given by

$$IG(Y|X) = H(Y) - H(Y|X)$$

That is, we can interpret $H(Y)$ as the uncertainty in Y, and $H(Y|X)$ as the expected uncertainty that would remain in Y after seeing X. Note that if X is completely uninformative about Y, then $IG(Y|X) = 0$. If X is completely informative about Y, then $IG(Y|X) = H(Y)$. The information gain of a split refers to how much information (over the training set) about the class label Y is gained by knowing which side of a split we are on.

A simple example illustrates this:

Let Y denote the random fruit ({oranges, lemons}), and let X denote the random side of the split ({left, right}).

Using the split on the left, $H(Y) = -\frac{2}{7}\log_2(\frac{2}{7}) - \frac{5}{7}\log_2(\frac{5}{7}) \approx 0.86$. $H(Y|\text{left}) \approx 0.81$, $H(Y|\text{right}) \approx 0.92$. $IG(\text{split}) = 0.86 - (\frac{4}{7} \cdot 0.81 + \frac{3}{7} \cdot 0.92) \approx 0.006$.

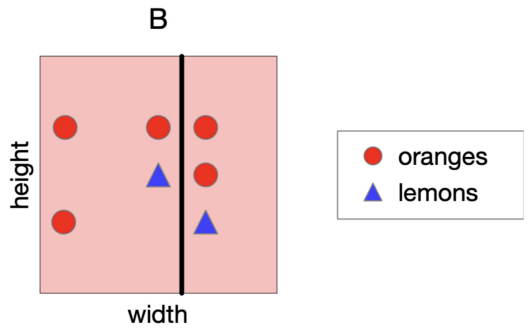

Figure 2: Splitting the training set based on width - first option

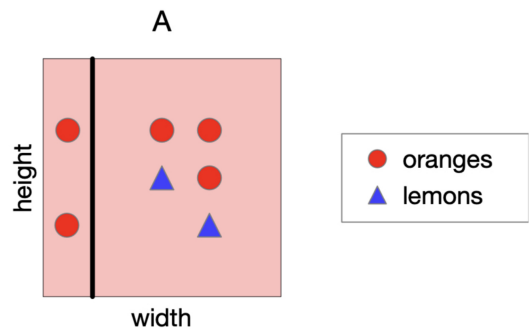

Figure 3: Splitting the training set based on width - second option

Using the split on the right, $H(Y) = H(Y|\text{right}) \approx 0.97$. $IG(\text{split}) = 0.86 - (\frac{2}{7} \cdot 0 + -\frac{2}{7}\log_2(\frac{2}{7}) - \frac{5}{7}\log_2(\frac{5}{7}) \approx 0.86$. $H(Y|\text{left}) = 0$, $\frac{5}{7} \cdot 0.97) \approx 0.17$.

As we can see the information gain of the second option is much higher the first. We therefore choose the split the second option as our split. Constructing a decision tree is similar in that at each level, the algorithm chooses the feature that gives the highest gain.

2.3 Decision Tree Construction

The decision tree construction uses a simple, greedy (i.e. makes locally optimal decisions at each step without considering the global optimal solution), recursive approach to build up tree node-by-node.

Algorithm (Decision Tree Construction):

1. Start with the entire dataset as the root node of the decision tree.
2. Calculate the information gain for all feature (including the ones used higher up in the tree) in the current node. Select the attribute with the highest information gain as the splitting criterion for the current node.
3. Create a branch from the current node for each possible value of the selected attribute. Divide the instances in the current node into subsets.
4. At the node,
   * If all the instances in the current node belong to the same class, create a leaf node with that class label.
   * If there are no more attributes to split on, create a leaf node with the majority class label of the instances in the current node.
   * Else, recursively apply the algorithm to each subset.
5. Return the decision tree.

An uncontrolled decision tree will grow indefinitely, and easily overfit the data. Therefore, we use hyperparameters, such as limiting the size of the training set, or the number of levels the tree could have, to best generalize the patterns in the data. An ideal decision tree is:

Not too small

* We need to handle important but possibly subtle distinctions in data

Not too big

* Computational efficiency (avoid redundant, spurious attributes)
* Avoid over-fitting training examples
* Human interpretability

When constructing a decision tree, ***Occam's razor*** suggests favoring simpler, more concise trees over complex ones. This means that when evaluating different splits at each node, the algorithm should prioritize attributes that provide the most significant information gain while keeping the tree as simple as possible.

Just as any machine learning algorithm, the decision tree has certain miscellanies. We may have exponentially less data at lower levels. Also, greedy algorithms don't necessarily yield the global optimum. When handling continuous attributes, we split based on a threshold, chosen to maximize information gain. Decision trees can also be used for regression on real-valued outputs. Choose splits to minimize squared error, rather than maximize information gain.

Although compared with k-NN, decision trees are simple to deal with discrete features, missing values, and poorly scaled data, fast at test time, and more interpretable, it uses more hyperparameters and and incorporate interesting distance measures (e.g. shape context).

3 Generalization and Bias-Variance Decomposition

Generalization and Bias-Variance Decomposition is one of the most important concepts in machine learning.

3.1 Generalization

When we train a machine learning model, we want it to *generalize* to data it hasn't seen before. Recall partitioning the data into three subsets:

Training set, a set of training examples the model is training on.

Validation set, a set used to tune hyperparameters such as the number of hidden units, or the learning rate.

Test set, a set used to measure the generalization performance.

Note the difference between a validation set and a test set: The test set is only used to report the final performance. We cannot tune hyperparamters on the test set, because that would be "cheating". Neither can we tune hyperparameters on the training set – that could lead us to choose a very expressive model that memorize the training data, which generalizes poorly.

There are two strategies for tuning hyperparamters: ***grid search*** and ***random search***, which differ in how the hyperparamters are chosen.

Grid search: For each hyperparameter, choose a set of candidate values. Separately train models using all possible combinations of these values.

Random search: Separately train models using combinations of random hyperparameter.

Then choose the configuration which gives the smallest validation error.. The advantage of random search over grid search is as follows:

Exploration of Hyperparameter Space: Random search allows for a more comprehensive exploration of the hyperparameter space compared to grid search. In grid search, the hyperparameter values are predefined and limited to a fixed set of values, which may miss important regions of the hyperparameter space. Random search randomly samples values for each hyperparameter from a specified range or distribution, enabling a wider exploration of the space.

Efficiency in High-Dimensional Spaces: As the number of hyperparameters increases, the grid search becomes computationally expensive since it requires evaluating all possible combinations of hyperparameter values. Random search only requires a fixed number of random samples, making it more efficient in high-dimensional spaces.

Handling Irrelevant Hyperparameters: In many cases, some hyperparameters have a minimal impact on the performance of the model. Grid search evaluates all possible combinations, including those with irrelevant hyperparameters, which wastes computational resources. Random search, by its random nature, may skip or assign lower priority to such irrelevant hyperparameters, focusing more on the ones that have a significant impact.

Let us now think about how the training and generalization error cary as a function of the number of training examples and the number of parameters.

Consider the number of training examples. The larger the training set, the more likely there will be a closely related training example, and the fewer the ***accidental regularities***, so the model will be forced to pick up the true regularities, so generalization error decreases. On the other hand, larger training sets are harder to memorize than small training sets, so training error increases.

Now consider the number of model capacity.[2] The more the parameters, the easier it is to fit both the accidental and the true regularities of the training data, so training error decreases. The effect on the generalization error is more subtle. If the model has too little capacity, it generalizes badly because it fails to pick up regularities, i.e. It underfits. if it has too much capacity, it will memorize the training set and fail to generalize, so the generalization error decreases then increases.

3.2 Bias-Variance Decomposition

Now we would like to mathematically model the generalization error. We introduce the ***data generating distribution***, a hypothetical distribution $p_{\mathcal{D}}(\mathbf{x}, t)$ that all the training and test data are assumed to have come from. We don't assume anything about the form of the distribution. The only nontrivial assumption is that the training and test data are drawn from the same distribution. A training set $\mathcal{D}$ consists of pairs $(\mathbf{x}_i, t_i)$ sampled ***independent and identically distributed (i.i.d.)*** from $p_{\mathcal{D}}(\mathbf{x}, t)$.

In quantifying the generlization error, what we would really like to decompose is the expec-

[2] Capacity isn't a formal term, but corresponds roughly to the number of trainable paramters.

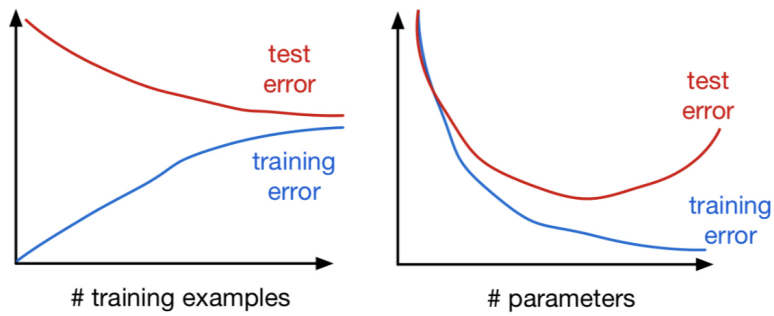

Figure 4: (left) Qualitative relationship between number of traning examples and training/test error. (right) Qualitative relationship between the number of parameters (or model capacity) and training/test error.

tation of the squared error loss,

$$L(y,t) = \frac{1}{2}(y-t)^2$$

where y is the prediction at a query point $\mathbf{x}$ and t is a sampled from a conditional distribution $p(t|x)$.

Basic setup (Bias-Variance Decomposition):

1. Fix a query point $\mathbf{x}$
2. Sample multiple training sets $\mathcal{D}$ i.i.d. from the data generating distribution $p_{\mathcal{D}}(\mathbf{x}, t)$
3. For each training set $\mathcal{D}$,
   * Run the learning algorithm on $\mathcal{D}$ to get a prediction y at $\mathbf{x}$
   * Sample a true target t from the conditional probability $p(t|x)$.
   * Compute the loss $L(x,t)$

This gives us a distribution over the loss at $\mathbf{x}$, with expectation $\mathbb{E}[L(y,t)|x]$. We would like to minimize this expectation.

Let us now consider only t to be a random variable (as it is sampled from a conditional probability), and y to be a constant (Consider only one). We claim that $y = \mathbb{E}[t|\mathbf{x}]$ is the best possible prediction. Here is a proof:

$$\begin{aligned}
\mathbb{E}[(y-t)^2|\mathbf{x}] &= \mathbb{E}[y^2 - 2yt + t^2|\mathbf{x}] \\
&= y^2 - 2y\mathbb{E}[t|\mathbf{x}] + \mathbb{E}[t^2|\mathbf{x}] \\
&= y^2 - 2y\mathbb{E}[t|\mathbf{x}] + (\mathbb{E}[t|\mathbf{x}])^2 + \mathrm{Var}[t|\mathbf{x}] \\
&= (y - \mathbb{E}[t|\mathbf{x}])^2 + \underbrace{\mathrm{Var}[t|x]}_{\text{Bayes error}}
\end{aligned}$$

This is minimized when $y = \mathbb{E}[t|\mathbf{x}]$. Notice that the ***Bayes error*** does not depend y. If the $\mathbb{E}[(y-t)^2|\mathbf{x}]$ is equal to the Bayes error, then the algorithm is ***Bayes optimal***. It is the best we can ever hope to do with any learning algorithm. The process of choosing y based on $p(t|\mathbf{x})$ is an example of decision theory.

Let us now treat y as a random variable, where the randomness comes from the choice of

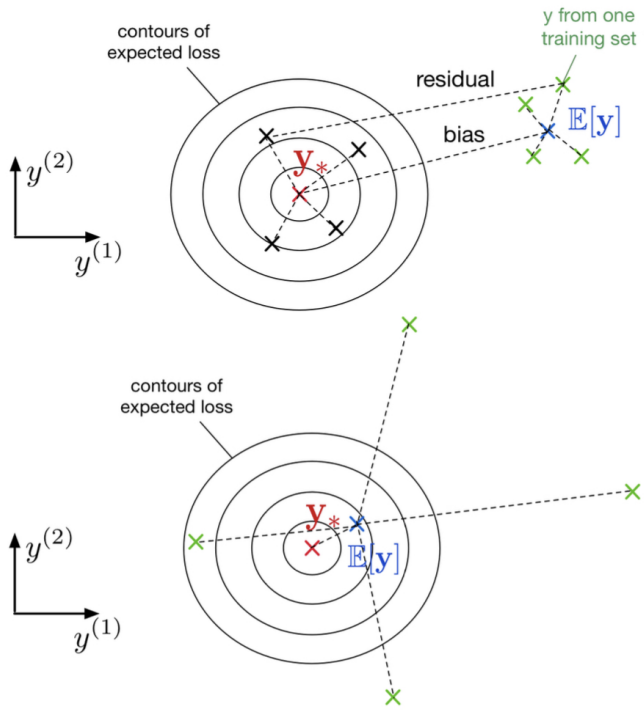

Figure 5: (top) If the model is underfitting, the bias is large but the variance is small. (bottom) If the model is overfitting, the bias is small but the variance is large

training set. Then we have

$$
\begin{aligned}
\mathbb{E}[(y-t)^2|\mathbf{x}] &= \mathbb{E}[y^2 - 2yt + t^2|\mathbf{x}] \\
&= \mathbb{E}[y^2|\mathbf{x}] - 2\mathbb{E}[yt|\mathbf{x}] + \mathbb{E}[t^2|\mathbf{x}] \\
&= (\mathbb{E}[y|\mathbf{x}])^2 + \mathrm{Var}[y|x] - 2\mathbb{E}[y|\mathbf{x}]\mathbb{E}[t|\mathbf{x}] + (\mathbb{E}[t|\mathbf{x}])^2 + \mathrm{Var}[t|x] \quad \text{Since } y \text{ and } t \text{ are independent} \\
&= \underbrace{(\mathbb{E}[t|\mathbf{x}] - \mathbb{E}[y|\mathbf{x}])^2}_{\text{bias}} + \underbrace{\mathrm{Var}[y|\mathbf{x}]}_{\text{variance}} + \underbrace{\mathrm{Var}[t|\mathbf{x}]}_{\text{Bayes error}}
\end{aligned}
$$

We have decomposed the expected loss into three terms, known as the ***bias-variance decomposition***:

* ***Bias***: How far off the model's average prediction is (corresponds to underfitting – The higher the bias, the worse the perfomance of the model)
* ***Variance***: Variability in the model's prediction across the training sets used (corresponds to overfitting – The higher the variance, the more the model overfits with the individual training sets)
* ***Bayes error***: The inherent unpredictability of the targets

3.3 Bagging

In short, bagging reduces over-fitting by averaging predictions.

Consider the errors of using one training set and multiple training sets. Suppose we sample N independent training sets from p_{sample}, and compute the prediction y_i using each training set, obtaining the average prediction $y = \frac{1}{m}\sum_{i=1}^{m} y_i$. Consider how this affects the bias and variance.

The bias is unchanged: $\mathbb{E}[y] = \mathbb{E}\left[\frac{1}{m}\sum_{i=1}^{m} y_i\right] = \mathbb{E}[y_i]$, since the averaged prediction has the same expectation. The variance is reduced: $\text{Var}[y] = \text{Var}\left[\frac{1}{N}\sum_{i=1}^{m} y_i\right] = \frac{1}{m^2}\sum_{i=1}^{m}\text{Var}[y_i] = \frac{1}{m}\text{Var}[y_i]$, since we are averaging over independent predictions.

Given training set $\mathcal{D}$, use the empirical distribution $p_{\mathcal{D}}$ as a proxy for p_{sample}. Note that as $|\mathcal{D}| \to \infty$, we have $p_{\mathcal{D}} \to p_{\text{sample}}$.

For binary classification (which we will discuss in a later section), we average the predictions and apply a threshold,

$$y_{\text{bagged}} = \mathbb{I}\left(\frac{1}{m}\sum_{i=1}^{m} y_i > 0.5\right)$$

just as majority vote.

Bagging / Bootstrap aggregation:

1. Take a training set $\mathcal{D}$ with n examples.
2. Generate n bootstrap samples, each containing n examples sampled from $\mathcal{D}$ with replacement.
3. Average the predictions of models training on the m dataset.

Random forests, one of the popular ensemble methods, are an extension of bagging where the base models are decision trees. Random forests combine bagging with random feature selection to further enhance the diversity and robustness of the ensemble models.

A bagged classifier can be stronger than the average model. It also reduces correlation between datasets by introducing additional variability. But note that if m datasets are not independent, we don't get the $\frac{1}{m}$ variance reduction. Bagging has certain limitations. It does not reduce bias in case of squared error. Also, weighting members equally my not be the best. Bagging can be sensitive to the specific training data samples. Since each base model is trained on a different bootstrap sample, the ensemble's performance can vary depending on the random sampling. In some cases, certain samples or outliers can have a disproportionate influence on the ensemble's predictions. Further, there could be high correlation between classifiers, leading to less variance reduction.

4 Linear models

The third learning algorithm we will look at is ***linear regression***. Linear regression exemplifies a modular approach that will be used throughout this set of notes:

* Choose a model decribing the relationship between variables of interest
* Define a ***loss function*** quantifying how bad the fit to the data is
* Choose a ***regularizer*** saying how much we prefer different candidate models

* Fit a model that minimizes the loss function and satisfies the constraint/penalty imposed by the regularizer, possibly using an ***optimization algorithm***.

4.1 Introduction

In linear regression, we use a linear function of the features $\mathbf{x} = (x_1, ..., x_D) \in \mathbb{R}^D$ to make predictions y of the target value $t \in \mathbb{R}$:

$$y = f(\mathbf{x}) = \sum_{j=1}^{D} w_j x_j + b$$

$$y = \mathbf{w}^\intercal \mathbf{x} + b$$

where y is the prediction, $\mathbf{w}$ is the weights, b is the bias.

The ***loss function*** $\mathcal{L}(y, t)$ quantifies the discrepancy between the predicted values and the true values. The squared error loss function is defined to be

$$\mathcal{L}(y, t) = \frac{1}{2}(y - t)^2$$

where $y - t$ is the ***residual*** which we would like to minimize, and the $\frac{1}{2}$ factor is for calculation convenience.

The ***cost function*** is the loss function averaged over all training examples and is defined to be

$$\begin{aligned}\mathcal{J}(\mathbf{w}, b) &= \frac{1}{2N} \sum_{i=1}^{N} \left(y^{(i)} - t^{(i)}\right)^2 \\ &= \frac{1}{2N} \sum_{i=1}^{N} \left(\mathbf{w}^\intercal \mathbf{x}^{(i)} + b - t^{(i)}\right)^2\end{aligned}$$

4.2 Vectorization

For our algorithm to be more efficient, we vectorize them by expressing them in terms of vectors and matrices. For the linear model $y = \mathbf{w}^\intercal \mathbf{x} + b$, it can be expressed in python simply by `y = np.dot(w, x) + b`.

We can organize all the training examples into a ***design matrix*** $\mathbf{X}$ with rows representing training examples, and columns representing features. We also organize the targets into the ***target vector*** $\mathbf{t}$

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}^{(1)\intercal} \\ \vdots \\ \mathbf{x}^{(N)\intercal} \end{pmatrix}$$

Computing the predictions for the whole dataset,

$$\mathbf{X}\mathbf{w} + b\mathbf{1} = \begin{pmatrix} \mathbf{w}^\intercal \mathbf{x}^{(1)} + b \\ \vdots \\ \mathbf{w}^\intercal \mathbf{x}^{(N)} + b \end{pmatrix} = \begin{pmatrix} y^{(1)} \\ \vdots \\ y^{(N)} \end{pmatrix} = \mathbf{y}$$

Computing the squared error cose across the whole dataset,

$$\mathbf{y} = \mathbf{X}\mathbf{w} + b\mathbf{1}$$
$$\mathcal{J} = \frac{1}{2N}||\mathbf{y} - \mathbf{t}||^2$$

We could also add a column of 1's to the design matrix and conveniently write

$$\mathbf{X} = \begin{bmatrix} 1 & [\mathbf{x}^{(1)}]^\intercal \\ \vdots & \vdots \\ 1 & [\mathbf{x}^{(N)}]^\intercal \end{bmatrix} \in \mathbb{R}^{N\times(D+1)} \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} b \\ w_1 \\ \vdots \\ w_D \end{bmatrix} \in \mathbb{R}^{D+1}$$

Then our predictions reduce to $\mathbf{y} = \mathbf{X}\mathbf{w}$.

4.3 Optimization

Our goal is to minimize the cost function $\mathcal{J}(\mathbf{x})$. Recall from calculus that the minimum of a smooth function, if it exists, occurs at a ***critical point***, i.e. where the derivative is zero.

$$\nabla_{\mathbf{w}}\mathcal{J} = \frac{\partial \mathcal{J}}{\partial \mathbf{w}} = \begin{pmatrix} \frac{\partial \mathcal{J}}{\partial w_1} \\ \vdots \\ \frac{\partial \mathcal{J}}{\partial w_D} \end{pmatrix}$$

Solutions may be direct or iterative.

* ***Direct solution***: Set the gradient to zero and solve in closed form – directly find provably optimal parameters
* ***Iterative solution***: Repeatedly apply an update rule that gradually takes us closed to the solution.

Let us look at the direct solution. We seek $\mathbf{w}$ to minimize $\mathcal{J}(\mathbf{w}) = \frac{1}{2}||\mathbf{X}\mathbf{w} - \mathbf{t}||^2$. From $||\mathbf{x}||_2 = \mathbf{x}^\intercal\mathbf{x}$ and properties of the transpose,[3]

$$\begin{aligned}
||\mathbf{X}\mathbf{w} - \mathbf{t}||^2 &= (\mathbf{X}\mathbf{w} - \mathbf{t})^\intercal(\mathbf{X}\mathbf{w} - \mathbf{t}) \\
&= ((\mathbf{X}\mathbf{w})^\intercal - \mathbf{t}^\intercal)(\mathbf{X}\mathbf{w} - \mathbf{t}) \\
&= (\mathbf{X}\mathbf{w})^\intercal\mathbf{X}\mathbf{w} - (\mathbf{X}\mathbf{w})^\intercal\mathbf{t} - \mathbf{t}^\intercal\mathbf{X}\mathbf{w} + \mathbf{t}^\intercal\mathbf{t} \\
&= \mathbf{w}^\intercal\mathbf{X}^\intercal\mathbf{X}\mathbf{w} - (\mathbf{X}\mathbf{w})^\intercal\mathbf{t} - \mathbf{t}^\intercal\mathbf{X}\mathbf{w} + \mathbf{t}^\intercal\mathbf{t} \\
&= \mathbf{w}^\intercal\mathbf{X}^\intercal\mathbf{X}\mathbf{w} - 2\mathbf{t}^\intercal\mathbf{X}\mathbf{w} + \mathbf{t}^\intercal\mathbf{t} \quad \text{since } \mathbf{X}\mathbf{w} \text{ and } \mathbf{t} \text{ are vectors} \\
&= \mathbf{w}^\intercal\mathbf{X}^\intercal\mathbf{X}\mathbf{w} - 2(\mathbf{X}^\intercal\mathbf{t})^\intercal\mathbf{w} + \mathbf{t}^\intercal\mathbf{t}
\end{aligned}$$

So then we have[4]

$$\begin{aligned}
\nabla(\frac{1}{2}||\mathbf{X}\mathbf{w} - \mathbf{t}||^2) &= \frac{1}{2}(-2\mathbf{X}^\intercal\mathbf{t} + (\mathbf{X}^\intercal\mathbf{X} + (\mathbf{X}^\intercal\mathbf{X})^\intercal)\mathbf{w}) \\
&= -\mathbf{X}^\intercal\mathbf{t} + \mathbf{X}^\intercal\mathbf{X}\mathbf{w} \quad \text{since } (\mathbf{X}^\intercal\mathbf{X})^\intercal = \mathbf{X}^\intercal\mathbf{X} \\
&= \mathbf{X}^\intercal\mathbf{X}\mathbf{w} - \mathbf{X}^\intercal\mathbf{t}
\end{aligned}$$

[3] For two matrices A and B, $(A+B)^\intercal = A^\intercal + B^\intercal$, $(AB)^\intercal = B^\intercal A^\intercal$

[4] See https://www.math.usm.edu/lambers/mat610/lecture8.pdf

Setting this to 0,

$$\begin{aligned}\mathbf{X}^\intercal\mathbf{X}\mathbf{w} &= \mathbf{X}^\intercal\mathbf{t} \\ \mathbf{w} &= (\mathbf{X}^\intercal\mathbf{X})^{-1}\mathbf{X}^\intercal\mathbf{t} \quad \text{since } \mathbf{X}^\intercal\mathbf{X} \text{ is invertible}\end{aligned}$$

However, many optimization problems do not have a direct solution. A more braodly applicable way to minimize the cost function is ***gradient descent***. Gradient descent is an ***iterative algorithm***, which means we apply an update repeatedly until some criterion is met. We initialize the weights to something reasonable, and repeatedly adjust them in the direction of steepest descent.

Notice that :

If $\partial\mathcal{J}/\partial w_i > 0$, then decreasing $\mathcal{J}$ requires decreasing w_i.

If $\partial\mathcal{J}/\partial w_i < 0$, then decreasing $\mathcal{J}$ requires increasing w_i

Consider a learning rate $\alpha > 0$. The larger α is, the faster $\mathbf{w}$ changes. If cost is the total loss rather than average loss, a smaller learning rate will be needed ($\alpha' = \alpha/N$). The following update always decreases the cost function for small enough α

$$w_i \leftarrow w_i - \alpha\frac{\partial\mathcal{J}}{\partial w_i}$$

Update rule in vector form:

$$\mathbf{w} \leftarrow \mathbf{w} - \alpha\frac{\partial\mathcal{J}}{\partial\mathbf{w}}$$

Update rule for linear regression:

$$\mathbf{w} \leftarrow \mathbf{w} - \frac{\alpha}{N}\sum_{i=1}^{N}(y^{(i)} - t^{(i)})\mathbf{x}^{(i)}$$

Gradient descent updates $\mathbf{w}$ in the direction of fastest decrease. Once it converges, there is a critical point at $\mathbf{w}$.

We use gradient descent because it is applicable to a much broader set of models, and easier to implement than direct solutions. For regression in high-dimensional space, gradient descent is more efficient than direct solution. For example, the linear regression direct solution $(\mathbf{X}^\intercal\mathbf{X})^{-1}\mathbf{X}^\intercal\mathbf{t}$ requires matrix inversion, which takes $\mathcal{O}(D^3)$. Each gradient descent update costs $\mathcal{O}(ND)$ or less with stochastic gradient descent. The difference is significant if $D >> 1$.

4.4 Feature Mapping

Consider mapping the input features to another space using $\psi(\mathbf{x}) : \mathbb{R}^D \to \mathbb{R}^d$, then treating the mapped feature (in $\mathbb{R}^d$) as the input of a linear regression procecdure. Then we can use linear regression to model a non-linear relationship.

Consider that we can fit the data using a degree-M polynomial function of the form

$$y = w_0 + w_1x + w_2x^2 + ... + w_Mx^M = \sum i = 0^M w_ix^i$$

The feature mapping is $\psi(x) = [1, x, x^2, ..., x^M]^\intercal$. $y = \psi(x)^\intercal\mathbf{w}$ is linear in $w_0, w_1, ..., w_M$. Hence we can use linear regression to find $\mathbf{w}$.

4.5 Regularization

The degree M of the polynomial controls the model's complexity. It is a hyperparameter for polynomial expansion and can be tuned using a validation set. Restricting the number of parameters / basis functions (M) is a crude approach to controlling the model complexity. Another approach is to keep the model large, but ***regularize*** it. A ***regularizer*** is a function that quantifies how much we prefer one hypothesis versus another.

A Probability

Consider the example of flipping a coin twice,

Sample space Ω is the set of all possible outcomes of an experiment.(e.g. $\Omega = \{HH, HT, TH, TT\}$)

Observations $\omega \in \Omega$ are points in the space, also called sample outcomes, realizations, or elements. (e.g. $\omega = HT \in \Omega$)

Events $E \subset \Omega$ are subsets of the sample space. (e.g. the event where both flips have the same outcome $E = \{HH, TT\} \subset \Omega$)

Axioms of probability The probability of an event $E, P(E)$, satisfies three axioms:

1. $P(E) \geq 0$ for every E
2. $P(\Omega) = 1$
3. If $E_1, E_2, ...$ are disjoint then
$$P(\bigcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} P(E_i)$$

Joint and conditional probability

$$P(A, B) = P(A|B)p(B) = P(B|A)P(A)$$

Independence Events A and B are independent if and only if $P(A, B) = P(A)P(B)$

Law of Total Probability

$$P(A) = \sum_B P(A, B) = \sum_B P(A|B)P(B)$$

Bayes' Rule

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \qquad \text{Posterior} = \frac{\text{Likelihood} * \text{Prior}}{\text{Evidence}}$$

Random variable A random variable is a mapping which assigns a real number $X(\omega)$ to each observed outcome $\omega \in \Omega$. A random variable is said to be independently and identically distributed (i.i.d.) if they are sampled from the same probability distribution and are mutually independent

Expectation value

$$\mathbb{E}[x] = \sum_{i=1}^{\infty} x_i p(x_i) \qquad \text{(univariate discreate R.V.)}$$

$$\mathbb{E}[x] = \int_{-\infty}^{\infty} x p(x) dx \qquad \text{(univariate continuous R.V.)}$$

$$\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y] \qquad \mathbb{E}[cX] = x\mathbb{E}[X]$$

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] \text{ only if } X \text{ and } Y \text{ are independent}$$

Variance

$$\mathrm{Var}[x] = \int_{-\infty}^{\infty} (x-\mu)^2 p(x) dx = \mathbb{E}[(x-\mu)^2] = \mathbb{E}[x^2] - (\mathbb{E}[x])^2$$

$$\mathrm{Var}[cX] = c^2 \mathrm{Var}[X]$$

$$\mathrm{Var}[X+Y] = \mathrm{Var}[X] + \mathrm{Var}[Y] \text{ only if } X \text{ and } Y \text{ are independent}$$

Discrete and Continuous Random Variable

Discrete Random Variables

* Takes countably many values
* Distribution defined by probablity mass function(PMF)
* Marginalization: $p(x) = \sum_y p(x,y)$

Continuous Random Variables

* Takes uncountably many values
* Distribution defined by probablity density function(PDF)
* Marginalization: $p(x) = \int_y p(x,y) dy$

B Linear Algebra

Norms In general, the l_p norm is defined to be

$$||x||_p = \left(\sum_i |x_i|^p \right)^{1/p}$$

The l_2-norm is called the Euclidean norm: $\sqrt{\sum_i x_i^2}$.

The l_1-norm is called the Manhattan norm: $\sum_i |x_i|$.

The l_{inf}-norm is called the max norm: $\max_i |x_i|$.